



(c) How to Conduct a Journal Club Presentation and Analyze a Paper

1. JOURNAL CLUBS #2 AND #3 DOCUMENTS

2A. THREE PRELIMINARY QUESTIONS

A. Three Preliminary Questions



Why was the study needed – and what was the research question?

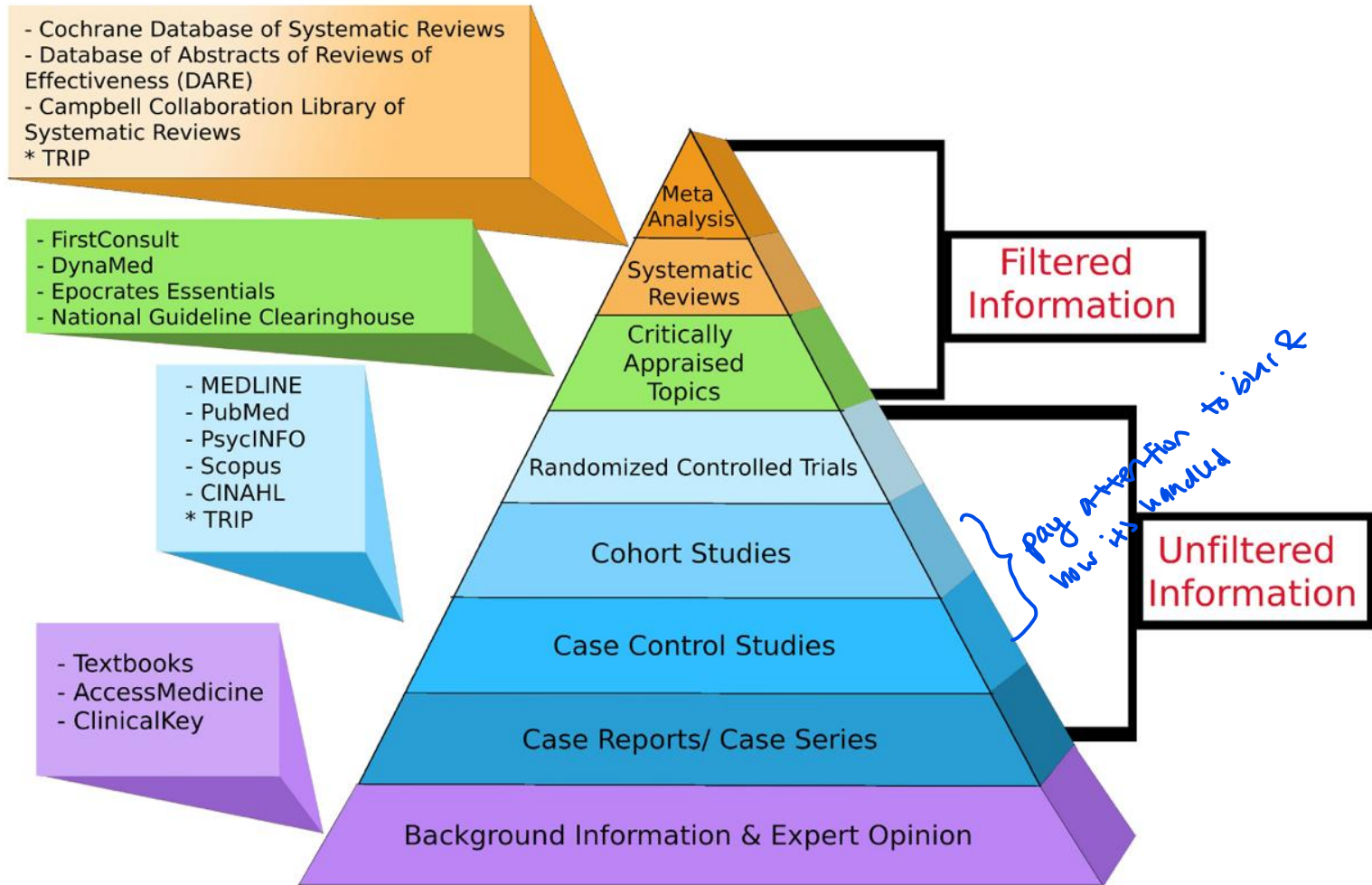


What was the research design?



Was the research design appropriate to answer the question?

B. Levels of Evidence Pyramid



*TRIP searches filtered AND unfiltered information simultaneously.

3. ASSESSING METHODOLOGICAL QUALITY

Assessing Methodological Quality

a

- Was the study original?

b

- Who was the study about?

c

- Was the design of the study sensible?

d

- Was bias minimized?

e

- Were preliminary statistical questions addressed?

be very clear on stats portion

4. STATISTICS/RESULTS

Statistics/Results

was the statistical test chosen correctly?

a. Have the authors set the scene correctly?

2 time points of one individual they are own control with time difference. Tails: one sided have a null hypothesis (do not know direction) or alt hypothesis (point to a specific direction)

see if have missing data to see if can trust the outcome

b. Paired data, tails, and outliers

correlation not equal association, retrospective observational = correlation (-1, 1) direction and quantification. Regression logistic (category and continuous will see odds ratio as outcome captured) and linear (continuous variables for both dependent and independent, change in every change in unit x). Causation: 5 criteria required (RCT intervention study controlling every factor between the two groups except for intervention)

c. Correlation, regression, and causation

can't use RCT for smoking and lung cancer - use lots of observational studies for causation, look at magnitude for measure of ci

interpret CI and knowing how to relay the message behind

d. Probability and confidence

e. The bottom line

Have the authors expressed the effects of an intervention in terms of the likely benefit or harm that an individual patient can expect?

5. APPLYING EVIDENCE TO PATIENT CARE

Decision-Guiding Aid

1. What is my risk of having a heart attack in the next 10 years?

NO STATIN

80 people DO NOT have a heart attack (green)

20 people DO have a heart attack (red)

The risk for 100 people like you who DO NOT take statins



YES STATIN

80 people still DO NOT have a heart attack (green)

5 people AVOIDED a heart attack (yellow)

15 people still DO have a heart attack (red)

95 people experienced NO BENEFIT from taking statins

The risk for 100 people like you who DO take statins



-  Had a heart attack
-  Avoided a heart attack
-  Didn't have a heart attack

2. What are the downsides of taking statins (cholesterol pill)?

- Statins need to be taken every day for a long time (maybe forever)
- Statins cost money (to you or your drug plan)
- Common side effects: nausea, diarrhoea, constipation (most patients can tolerate)
- Muscle aching/stiffness: 5 in 100 patients (some need to stop statins because of this)
- Liver blood test goes up (no pain, no permanent liver damage): 2 in 100 patients (some need to stop statins because of this)
- Muscle and kidney damage: 1 in 20 000 patients (requires patients to stop statins)

3. What do you want to do now?

- ☐ Take (or continue to take) statins
- ☐ Not take (or stop taking) statins
- ☐ Prefer to decide at some other time

Questions?